

Properties of movement

Animal tracking 25/26

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- ▶ In previous classes we have discussed about types of spatial data, we looked really broadly about the drivers of species distribution, and we looked at calculating home ranges using data accessed from the Movebank website
- ▶ Here we try to look more in details about geometric properties of animal movement data, without taking too much in to account the environment

- ▶ We have already formalized movement as a series of relocations over time
- ▶ Each point in the tracked movement of an individual can be defined by a time stamp (t) and by coordinates collected within a defined reference system (x and y)

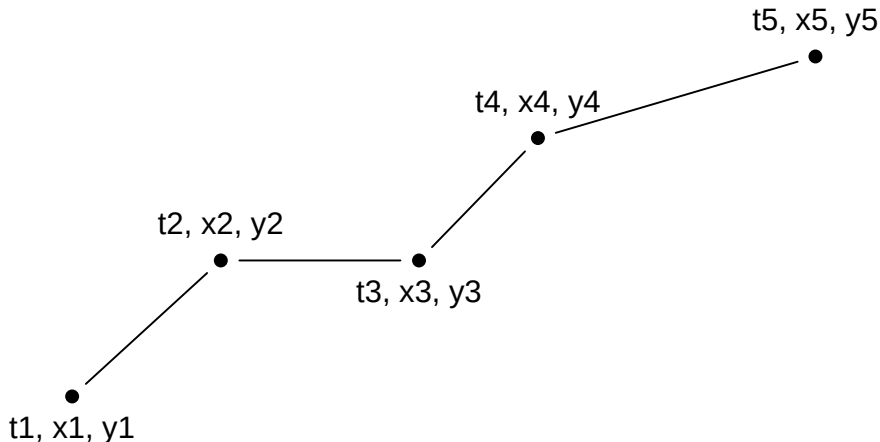


Figure 1: Tracked movement of an individual. Each point of the track is characterized by a timestamp (t) and by a set of coordinates (x and y)

- ▶ We can now describe the property of this track by looking at some of its features like:
 - ▶ Time lag between re-locations
 - ▶ Distance
 - ▶ Speed
 - ▶ Azimuth (aka, direction of movement)
 - ▶ Turning angles

- ▶ If we start with a sequence of timestamps along a trajectory, the time lag of a track is defined as the time difference between one step and the following step
- ▶ Time lag is a property of the track and not of the points
- ▶ If we have n points we can calculate only $n-1$ time lags. One missing value is usually automatically added at the end as you can not calculate the time lag of the last trajectory relocation
- ▶ We differentiate between a scheduled time lag and an operational time lag. The first one is what the time lag should be assuming the device is collecting data in the exact way it was programmed for. The operational time lag is the actual calculated time lag

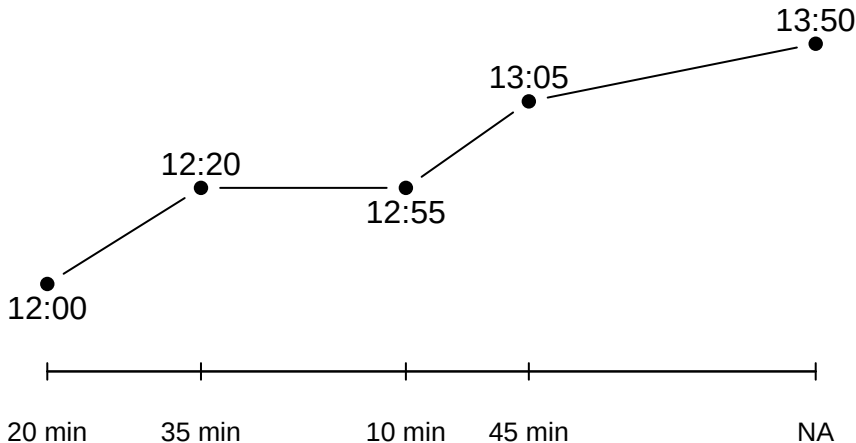


Figure 2: Time lag between relocations. The time difference between the first time stamp and the second time stamp is 20 minutes. The difference between the second and third is 35 minutes...and so on and so forth!

- ▶ If we look at coordinates of points along a trajectory we can measure the distance between each consecutive point. Assuming we are collecting coordinate every step of the individual we can call this property of the segment step length
- ▶ Distance is a property of the track and not of the points
- ▶ If we have n points we can calculate only $n-1$ distances. One missing value is usually automatically added at the end as you can not calculate distance of the last trajectory relocation

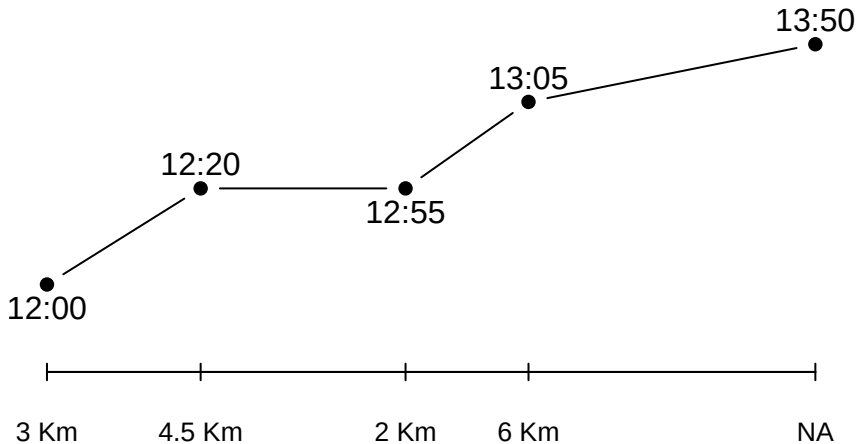


Figure 3: Distance between relocations.

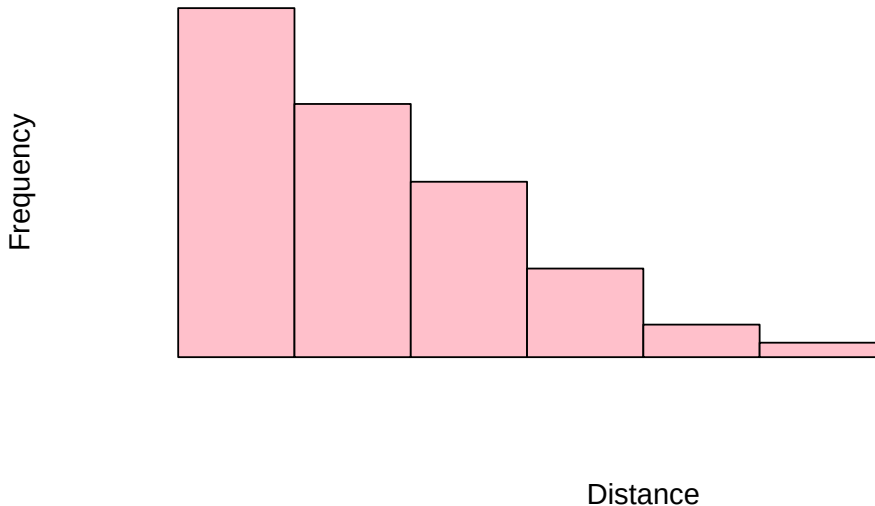


Figure 4: Typical distribution of step length distances. This is what we would expect to see as in most cases individuals would cover short distances at each step and in more rarely move faster and thus cover longer distances

- ▶ Speed (often referred as ground speed) is calculated as step length divided by time lag ($\frac{Step\ length}{Time\ lag}$)
- ▶ Speed as well is a property of the track and not of the points

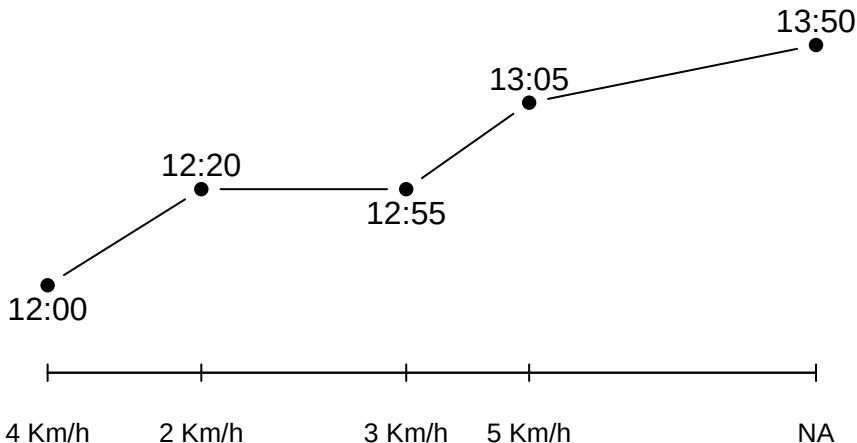


Figure 5: Ground speed between relocations. We would expect a similar distribution of values than the distance distribution. Most animals tend to move at their normal speed and sometimes they move faster if there is a reason to do.

- ▶ The angle formed between a reference direction (usually the North) and a line from the observer to a point of interest (the next relocation)
- ▶ Azimuth as well is a property of the track and not of the points
- ▶ When talking about movements we should make a distinction between heading and direction! The heading refers to the orientation of the body (or the axes of the body) during the movement, the direction is the actual direction of the movement. A crab may be heading north (its head and eyes may look to the north), but its direction of movement is usually lateral so would be moved east or west!

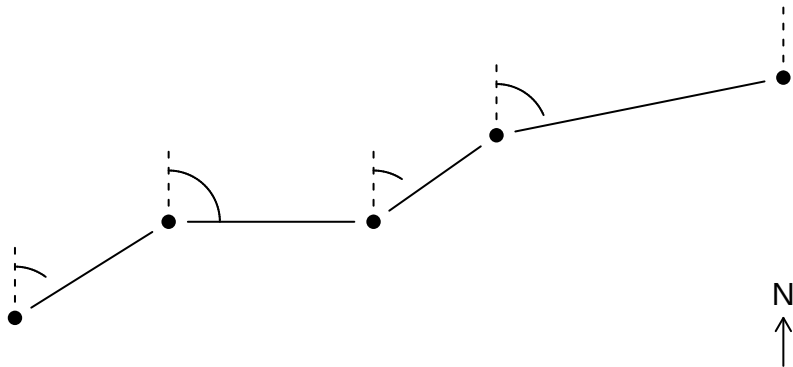
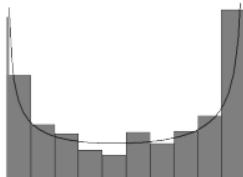


Figure 6: Azimuth. The dash segments represent our reference line (the north) and the angle between that and our segment is the azimuth.

Typical distribution



Local movement



0 360

North migration

Figure 7: Distribution of azimuth values when animals are roaming around randomly (top graph) or if the animals are moving along a geographic feature (for example along a river) or migrating. Image from the AniMove series of lessons.

- ▶ Relative angle between consecutive segments
- ▶ It is as well a property of the track but because we work with segments we need to exclude two points from the total number of relocations
- ▶ Left turn: negative values; right turn: positive values
- ▶ The angle values go between -180 and $+180$ degrees OR $-\pi$ to π (or π to 2π)

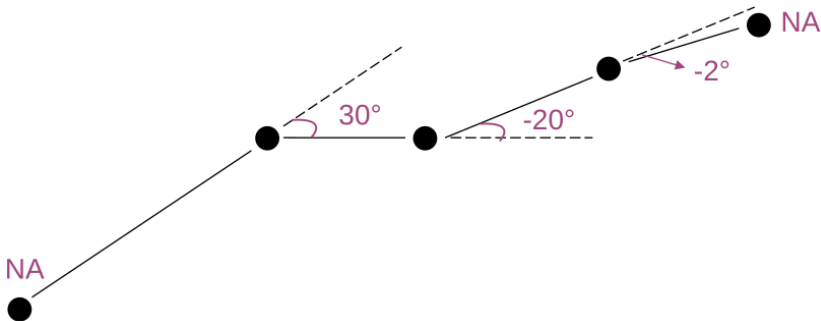


Figure 8: Turning angles. Image from the AniMove series of lessons.

Typical distribution

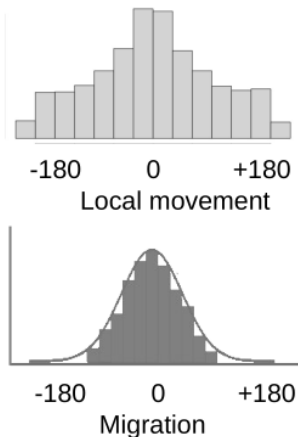


Figure 9: Distribution of turning angles in local movements and during migrations. Image from the AniMove series of lessons.

Effect of sampling frequency

Most trajectories have **gaps and shifts**.

This affects the **calculation and comparability of movement metrics**.

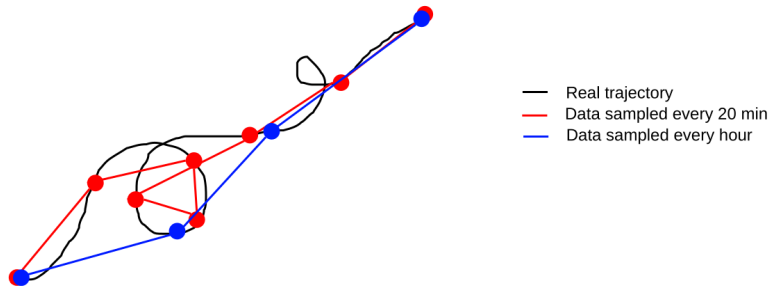


Figure 10: Effect of sampling frequency on the estimated trajectory of an individual. Image from the AniMove series of lessons.

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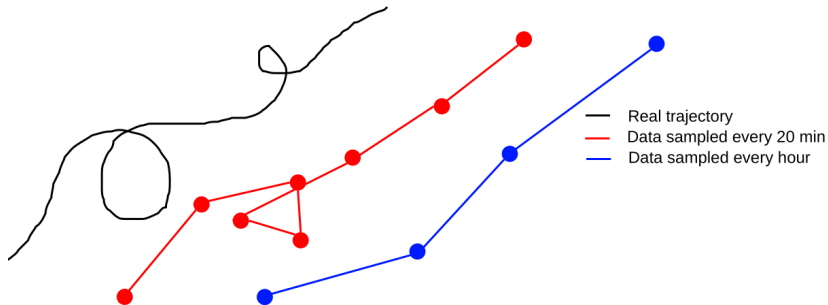


Figure 11: Effect of sampling frequency on the estimated trajectory of an individual. Image from the AniMove series of lessons.

- ▶ Interpolation: (fill locations in gapy sections). Problem: underestimating speed, variance in azimuth and turning angle. Introduces artificial autocorrelation in the data. Interpolation has to be used with care!
- ▶ Subsampling/thinning: (drop locations to reduce data to largest gap in time). Problem: dropping data removes auto-correlation and affects (underestimates) speed and turning angle.
- ▶ Segment and work with different parts of your data depending on the purpose.